

Mohamad Ali Al Saidi

Machine Learning Engineer | Computer Vision | Edge AI | LLMs

Beirut, Lebanon | +961 3 013 113 | Eng.mohamad.ali.saidi@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Machine Learning Engineer and Ph.D. candidate in Artificial Intelligence and Robotics with a strong software engineering foundation and hands-on experience across computer vision, multilingual NLP, large language models, model compression, and Edge AI.

Experienced in developing ML systems from data preparation, model training, evaluation, and optimization through application integration and deployment across desktop, local-inference, and resource-constrained embedded environments.

Strong background in Python, PyTorch, TensorFlow, Java, and SQL, combining applied machine learning with practical experience developing and maintaining production software systems.

Technical Skills

Machine Learning & Deep Learning: Python, PyTorch, TensorFlow, Keras, Scikit-learn, neural network training, evaluation, hyperparameter tuning, and experiment analysis

NLP & Large Language Models: Hugging Face Transformers, BERT, T5, Llama, Mistral, Qwen, NLLB, M2M100, tokenization, embeddings, translation, summarization, RAG, and local LLM inference

Computer Vision & Multimodal AI: OpenCV, OCR, object detection, segmentation, YOLOv5, YOLOv8, R-CNN, Fast R-CNN, Faster R-CNN, Mask R-CNN, CLIP, BLIP, and LLaVA

Model Optimization & Edge AI: Knowledge distillation, model pruning, quantization, TensorFlow Lite, TensorFlow Lite Micro, TinyML, and Arduino Nano 33 BLE

Software Engineering: Java, C/C++, SQL, Git, Linux, Docker, Bash, PowerShell, and LaTeX

Selected Projects

Passport Field Detection and OCR | YOLOv8, EasyOCR, Tesseract, OpenCV, Python

- Trained a YOLOv8n model on a custom dataset of 604 passport images to detect seven structured fields: MRZ, first name, last name, passport number, birth date, expiry date, and gender.
- Achieved 92.76% mAP@0.5 and 94.26% recall and integrated field detection with EasyOCR, Tesseract, MRZ check-digit correction, structured data extraction, and a desktop application.

Model Pruning and Knowledge Distillation | PyTorch, VGG19, CIFAR-100, Model Compression

- Reproduced and extended a VGG19 compression pipeline on CIFAR-100 using iterative unstructured pruning, recovery training, and knowledge distillation.
- Achieved 39.47% sparsity in the pruned teacher and reduced the student from 40.18M to 16.37M parameters and from 153.35 MB to 62.48 MB while maintaining 73.20% accuracy.

TinyML Gesture Recognition on Arduino Nano 33 BLE | TensorFlow, CNN, LSTM, TensorFlow Lite Micro, Arduino

- Collected a custom flex-and-wave gesture dataset using six-axis accelerometer and gyroscope measurements from an Arduino Nano 33 BLE.
- Developed CNN and LSTM pipelines for two-class, six-feature time-series classification, converted and quantized the trained model using TensorFlow Lite, and deployed it for real-time microcontroller inference.

Fine-Tuned Local Multilingual Translator | Transformers, NLLB, M2M100, PyTorch, Python

- Fine-tuned multilingual transformer models on a private dataset for Persian, Arabic, and English translation and produced updated model weights for domain-specific local inference.
- Developed an offline desktop application with text segmentation, structural and newline preservation, decoding improvements, model caching, and privacy-preserving local inference.

Professional Experience

Airline System Developer

2018 – 2025

Sky Gift

Beirut, Lebanon

- Developed and maintained a modular airline reservation and operations platform supporting passenger records, booking workflows, ticketing, flight-related information, and internal operational processes using Java, SQL, and related application technologies.
- Integrated a computer-vision and OCR workflow into passenger-data processing, combining passport field detection, OCR, MRZ validation, and structured extraction to reduce manual handling of passport information.
- Developed Python and SQL utilities for data extraction, transformation, validation, reporting, and structured information exchange between operational components and internal workflows.
- Built and maintained desktop interfaces and operational tools that provided staff with structured access to booking, passenger, flight, and administrative information while supporting the reliability of the underlying databases and application environment.

System Administrator Intern

2017 – 2018

ICT Department, Ferdowsi University of Mashhad

Mashhad, Iran

- Supported Windows and Linux systems, servers, user accounts, network equipment, operating-system maintenance, backups, and technical troubleshooting across the university ICT environment.
- Automated recurring system-administration and maintenance tasks using Bash and PowerShell scripts.

Teaching & Training Experience

Machine Learning Course Co-Instructor

Spring 2026

Ferdowsi University of Mashhad

Mashhad, Iran

- Co-taught a semester-long machine learning course with academic supervisor, sharing responsibility for course delivery throughout the semester.

AI in Education Workshop Instructor

February 2026

Hamrah Aval Academy

Baghdad & Karbala, Iraq

- Delivered a two-day live workshop series for teachers on the practical and responsible use of AI in education, covering prompt design, lesson planning, assessment materials, presentations, and classroom content creation.

Education

Ph.D. Candidate in Computer Engineering, Artificial Intelligence & Robotics

2023 – Present

Ferdowsi University of Mashhad

Mashhad, Iran

Doctor of Business Administration (DBA)

2024 – 2026

University of Tehran

Tehran, Iran

Master of Business Administration (MBA)

2022 – 2023

Ferdowsi University of Mashhad

Mashhad, Iran

M.Sc. in Computer Engineering, Artificial Intelligence & Robotics

2020 – 2022

Ferdowsi University of Mashhad | GPA: 3.9/4.0

Mashhad, Iran

- *Thesis:* Hierarchical Solving of Euclidean Geometry Ruler-and-Compass Construction Problems Using Visual Knowledge Representation and Curriculum Learning.

B.Sc. in Computer Engineering, Artificial Intelligence

2016 – 2020

Ferdowsi University of Mashhad | GPA: 3.9/4.0

Mashhad, Iran

Languages

Arabic: Native | English: Fluent (C1) | Persian: Fluent (C2)